Loi Binomiale

DISCRET

ser un tampon pour
er le tableau
et non
évues exclusivement
ivage des tables.

- Lancer un dé est une E.A à 2 issues :

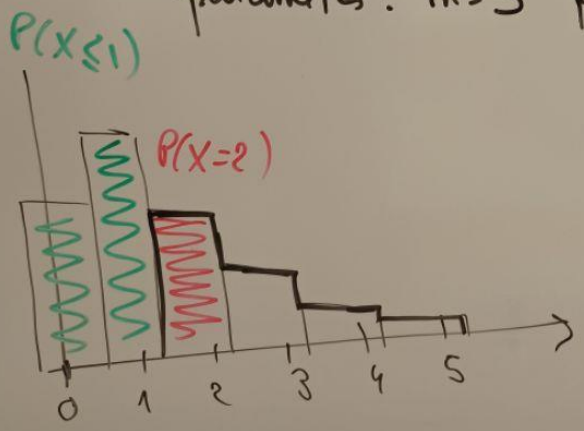
S = "obtenir un 6" $p = \frac{1}{6}$

$\bar{S}$: "ne pas obtenir un 6" $1-p = \frac{5}{6}$

- On répète $n=5$ fois cette EA de manière ID et IND.

- X est la V.A qui compte le nombre de succès.

- Donc X suit la loi Binomiale de paramètres : $n=5$ $p=\frac{1}{6}$ $X \sim B(5;\frac{1}{6})$



$E(X) = np$
$= 5 \times \frac{1}{6} = \frac{5}{6}$

$\sigma(X) = \sqrt{np(1-p)}$

$P(X=k)$: Binomfdp $\begin{cases} \text{nbre Essais} : n = 5 \\ p : p = \frac{1}{6} \\ \text{valeur de } x : k \end{cases}$

$P(X=2)$ = Binomfdp $\left(5, \frac{1}{6}, 2\right) = 0{,}16$

$P(X \leq k)$: Binomfrep $\begin{cases} \text{nb Essais} : n = 5 \\ p : p = \frac{1}{6} \\ \text{Valeur de } x : k \end{cases}$

$P(X \leq 1)$ = Binomrep $\left(5; \frac{1}{6}, 1\right)$

$P(X \geq k)$ $P(X \geq 2)$ = NOIR − ROUGE $= 1 - P(X \leq 1)$

~~0~~	~~1~~	2	3	4	5

$= 1 -$ Binomfrep $\left(5; \frac{1}{6}; 1\right.$

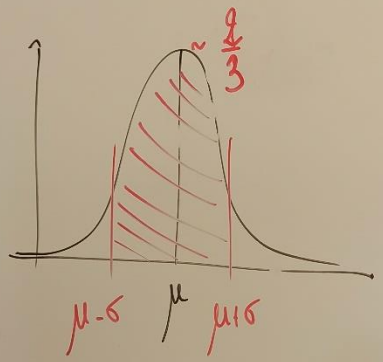
$= 0{,}19$

LOI NORMALE (CONTINUE)

Y suit la loi normale de paramètre μ et σ

Moyenne — Ecart type.

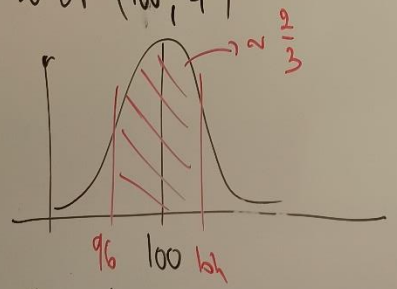
$Y \sim \mathcal{N}(\mu ; \sigma)$



$P(Y = k) = 0$

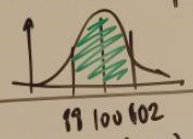
$P(a \leq Y \leq b) = \text{NormalFrep}\begin{cases} \text{Binf} : a \\ \text{Bsup} : b \\ \mu : \mu \\ \sigma : \sigma \end{cases}$

$Y \sim \mathcal{N}(100 ; 4)$

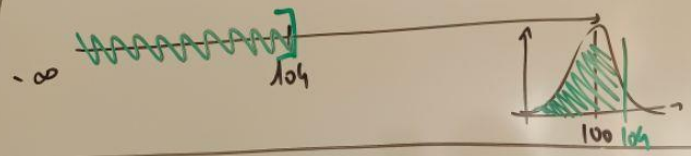


$P(88 \leq Y \leq 102) = \text{NormalFrep}(88, 102, 100, 4) = 0.69$

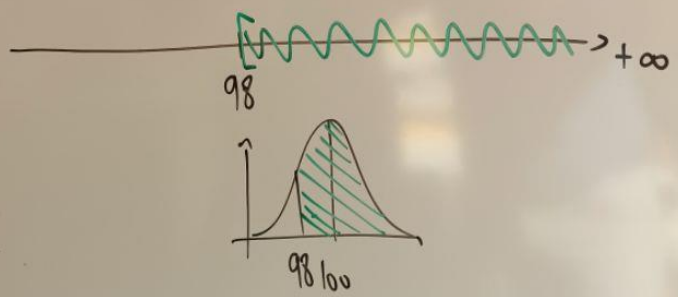
($88 \to a$, $102 \to b$)



$P(Y \leq b)$ $\quad P(Y \leq 104) = P(-10^{99} \leq Y \leq 104) = 0.84$

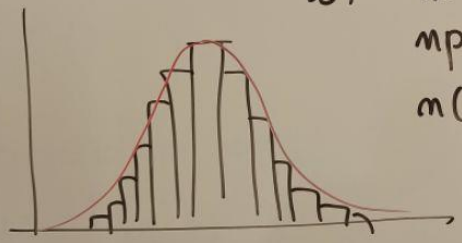


$P(Y \geq a)$ $\quad P(Y \geq 98) = P(98 \leq Y \leq 10^{99}) = 0.69$



$X \sim \mathcal{B}(n, p)$

On va approcher X par $Y \sim \mathcal{N}(\mu, \sigma)$



Si $n > 30$

$np > 5$

$n(1-p) > 5$

$\longrightarrow$ $\mu = n \times p$

$\sigma = \sqrt{np(1-p)}$